

# AUA MONOPOLY

*UNIQUE EDITION*

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Java implementation of the famous board game Monopoly.



# THE IDEA



## 40 Tiles

Water Works, Jail, Cascade Complex, etc.

## 22 Properties

8 color groups — KFC (\$60) to MIT (\$400)

## 4 Metro Stations

Replacing railroads — Republic Square, Zoravar Andranik & more

## 50 Unique Cards

Including Chance & Community Chest

# THE BOARD

|                           |                                                                                                     |                |               |                      |                             |                        |                        |                      |                            |                                                                              |
|---------------------------|-----------------------------------------------------------------------------------------------------|----------------|---------------|----------------------|-----------------------------|------------------------|------------------------|----------------------|----------------------------|------------------------------------------------------------------------------|
| JAIL                      | Chanel<br>\$220                                                                                     | Chance         | Dior<br>\$220 | Gucci<br>\$240       | Republic Square<br>\$200    | Yerevan Hotel<br>\$260 | Dilijan Hotel<br>\$260 | Water Works<br>\$150 | Tsaghkadzor Hotel<br>\$280 | Republic Square                                                              |
| Mega Mall<br>\$200        | <div>AUA MONOPOLY</div> <div>Unique Edition</div> <div>Alice's turn</div> <div>?</div> <div>?</div> |                |               |                      |                             |                        |                        |                      |                            | Paris<br>\$300                                                               |
| Dalma Mall<br>\$180       |                                                                                                     |                |               |                      |                             |                        |                        |                      |                            | New York<br>\$300                                                            |
| Community Chest           |                                                                                                     |                |               |                      |                             |                        |                        |                      |                            | Community Chest                                                              |
| Yerevan Mall<br>\$180     |                                                                                                     |                |               |                      |                             |                        |                        |                      |                            | Oxford University<br>\$320                                                   |
| Yeritsardakan<br>\$200    |                                                                                                     |                |               |                      |                             |                        |                        |                      |                            | Zoravar Andranik<br>\$200                                                    |
| Pull & Bear<br>\$160      |                                                                                                     |                |               |                      |                             |                        |                        |                      |                            | Chance                                                                       |
| Bersika<br>\$140          |                                                                                                     |                |               |                      |                             |                        |                        |                      |                            | Harvard University<br>\$350                                                  |
| Electric Company<br>\$150 |                                                                                                     |                |               |                      |                             |                        |                        |                      |                            | Luxury Tax<br>\$100                                                          |
| Zara<br>\$140             |                                                                                                     |                |               |                      |                             |                        |                        |                      |                            | MIT University<br>\$400                                                      |
| Cascade Complex           | Gotcha<br>\$120                                                                                     | Cofix<br>\$100 | Chance        | Coffe House<br>\$100 | Marshal Baghramyan<br>\$200 | Income Tax<br>\$200    | McDonald's<br>\$60     | Community Chest      | KFC<br>\$60                | GO<br><div><div></div><div>D</div><div>C</div><div>B</div><div>A</div></div> |



## GO — Bottom Right

Pass from the second cycle the GO, collect \$200



## 8 different property categories

From KFC to MIT  
— cheapest → most expensive



## 4 Metro Stations

Marshal Baghramyan → Zoravar Andranik



## Four Tokens

For playing with the whole friendgroup

# CLASS ARCHITECTURE

16 Classes

## GAME LOGIC

Game

Player

Board

Dice

Bank

CardDeck

Tile

## TILE HIERARCHY

« abstract »  
Tile

Property

MetroStation

GoTile

TaxTile

JailTile

FreeTile

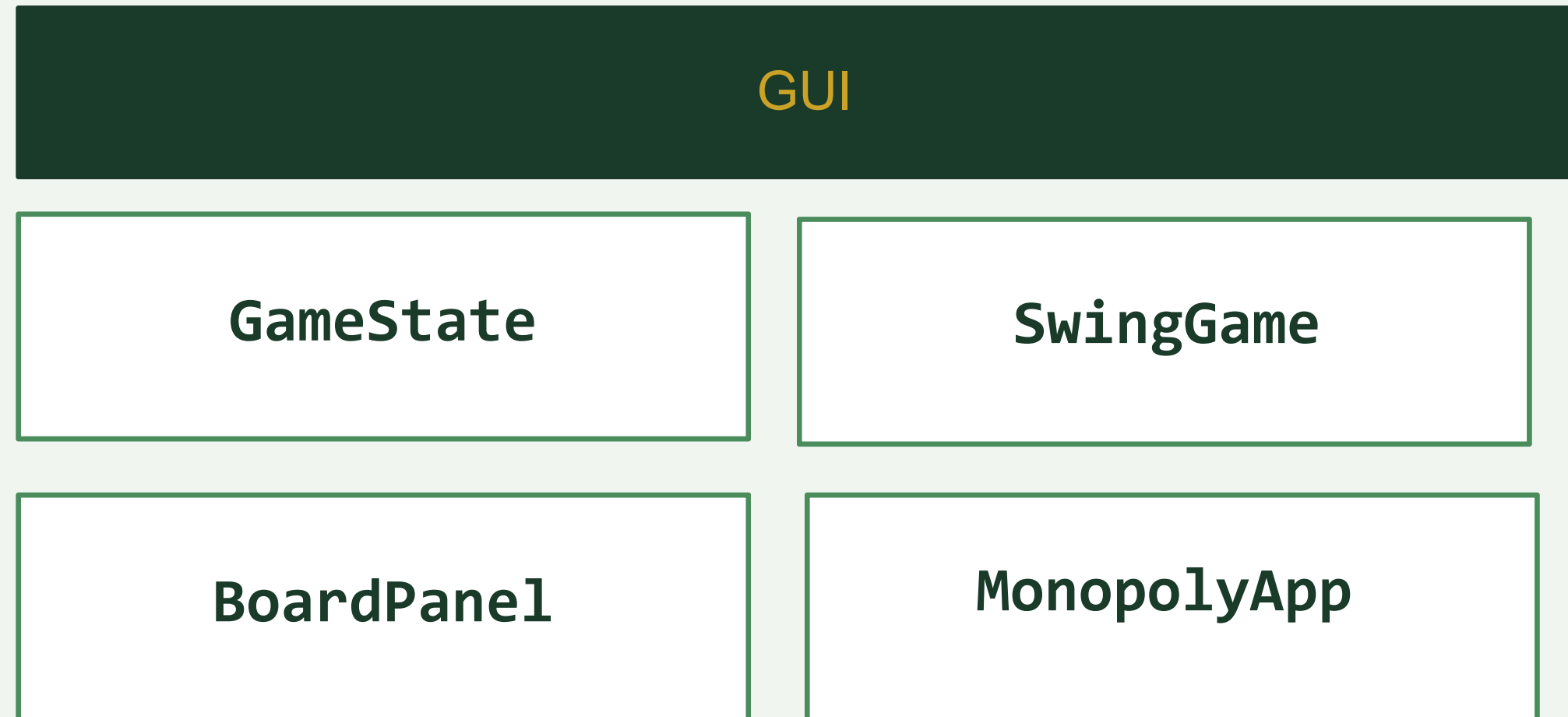
ChanceTile

CommunityTile

Game.java:    `player.move()`    →    `tile.landOn(player)`    →    correct behavior fires automatically

# CLASS INTERFACE

4 Classes





# OOP CONCEPTS

## ◆ ABSTRACTION

Tile defines what every tile must do — subclasses decide how.

```
abstract void landOn(Player p);
```

## ◆ POLYMORPHISM

Two lines handle all 40 tiles differently. Eight outcomes, zero if-else chains.

```
tile = board.getTile(pos);  
tile.landOn(player);
```

## ● INHERITANCE

All 8 tile types extend Tile — zero duplicated code.

```
class Property extends Tile { ... }
```

## ◎ ENCAPSULATION

No tile touches money directly — all transactions go through Bank.

```
bank.transfer(from, to, amount);
```

# THE BANK



## \$20,580

Exact real Monopoly amount

01

### Dedicated Class

No shortcuts like `player.money += 200`

02

### All Transactions

Every dollar in and out goes through Bank

03

### Single Source of Truth

One place for all money logic

04

### Bank Can Run Out

Just like real Monopoly

# CARDS

50

*unique  
cards*

## CHANCE

*Advance to  
Yeritasardakan Station.*

## COMMUNITY CHEST

*Bank pays you  
dividend of \$50.*

## CHANCE

*Take a trip to  
Republic Square.*

## COMMUNITY CHEST

*You won a beauty  
contest — collect \$10.*

Each card: 2 fields — description (String) + cashEffect (int)



# CUSTOM JAIL RULE

*If four players land in jail simultaneously — all go free.*

*Java Streams Implementation*

```
if (activeInJail == activeTotal && activeTotal == 4) {  
    System.out.println("All 4 players are in Jail! Everyone is released.");  
    allPlayers.stream()  
        .filter(Player::isInJail)  
        .forEach(Player::releaseFromJail);  
}
```

**filter()**

Select only jailed players

**count()**

Check how many are in jail

**forEach()**

Set each one free

# PROPERTIES

*8 Color Groups*

|       |            |       |        |       |        |       |           |
|-------|------------|-------|--------|-------|--------|-------|-----------|
| \$60  | \$100      | \$140 | \$180  | \$220 | \$260  | \$300 | \$400     |
| Brown | Light Blue | Pink  | Orange | Red   | Yellow | Green | Dark Blue |

## THREE LINES OF LOGIC — EVERY LANDING RESOLVED

### 1 No Owner

*Offer to buy*

```
if (!prop.isOwned()) {  
  if (p.getMoney() >=  
prop.getPrice()) {  
    state.pendingBuy = prop;  
    state.phase =  
GameState.Phase.BUY;  
  }  
}
```

### 2 Someone Else Owns It

*Pay rent*

```
else {  
  Player owner = prop.getOwner();  
  p.subtractMoney(prop.getRent());  
  owner.addMoney(prop.getRent());  
  state.addLog("🏠 " + p.getName() + "  
pays $" + prop.getRent() + " rent to " +  
owner.getName() + " for " +  
prop.getName());  
  checkBankruptAndContinue(p);  
}
```

### 3 You Own It

*You're told*

```
else if  
(prop.getOwner().equals(p)) {  
  state.addLog("🏠 " + p.getName()  
+ " owns " + prop.getName()  
+ ". No rent due.");  
  
  finishTurn(p);  
}
```

*Now let's show you the  
game.*

## Seda & Anna

AUA Monopoly — Unique Edition | Built in Java, OOP

40 Tiles

22 Properties

20 Classes

50 Cards

4 Metros